

## ✓ Python Data Analytics Class Exercise

### Grocery/Retail Data Analysis — SQL Querying & Data Analysis

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#### Exercise Description

This document presents my completed class exercise using grocery/retail data. The exercise demonstrates the application of SQL querying, data aggregation, filtering, inventory turnover analysis, and cross-tabulation to examine structured retail supply chain and sales data.

#### ✓ 1. Import the libraries

```
import pandas as pd
```

#### ✓ 2. Import the data from two csv files as DataFrames `grocery\_purchase` and `grocery\_transaction`)

```
grocery_purchase=pd.read_csv('QVI_purchase_behaviour.csv')
```

```
grocery_transaction=pd.read_csv('QVI_transaction_data.csv')
```

#### ✓ 3. Look at the info summary, head of each DataFrame

```
grocery_purchase.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 72637 entries, 0 to 72636
Data columns (total 3 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   LYLTY_CARD_NBR   72637 non-null  int64  
1   LIFESTAGE        72637 non-null  object  
2   PREMIUM_CUSTOMER 72637 non-null  object  
dtypes: int64(1), object(2)
memory usage: 1.7+ MB
```

```
grocery_purchase.head()
```

|   | LYLTY_CARD_NBR | LIFESTAGE              | PREMIUM_CUSTOMER |
|---|----------------|------------------------|------------------|
| 0 | 1000           | YOUNG SINGLES/COUPLES  | Premium          |
| 1 | 1002           | YOUNG SINGLES/COUPLES  | Mainstream       |
| 2 | 1003           | YOUNG FAMILIES         | Budget           |
| 3 | 1004           | OLDER SINGLES/COUPLES  | Mainstream       |
| 4 | 1005           | MIDAGE SINGLES/COUPLES | Mainstream       |

```
grocery_transaction.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 264836 entries, 0 to 264835
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0    DATE            264836 non-null  int64
1    STORE_NBR       264836 non-null  int64
2    LYLTY_CARD_NBR  264836 non-null  int64
3    TXN_ID          264836 non-null  int64
4    PROD_NBR        264836 non-null  int64
5    PROD_NAME       264836 non-null  object
6    PROD_QTY        264836 non-null  int64
7    TOT_SALES       264836 non-null  float64
dtypes: float64(1), int64(6), object(1)
memory usage: 16.2+ MB
```

```
grocery_transaction.head()
```

|   | DATE  | STORE_NBR | LYLTY_CARD_NBR | TXN_ID | PROD_NBR | PROD_NAME                                | PROD_QTY | TOT_SALES |
|---|-------|-----------|----------------|--------|----------|------------------------------------------|----------|-----------|
| 0 | 43390 | 1         | 1000           | 1      | 5        | Natural Chip Compny SeaSalt175g          | 2        | 6.0       |
| 1 | 43599 | 1         | 1307           | 348    | 66       | CCs Nacho Cheese 175g                    | 3        | 6.3       |
| 2 | 43605 | 1         | 1343           | 383    | 61       | Smiths Crinkle Cut Chips Chicken 170g    | 2        | 2.9       |
| 3 | 43329 | 2         | 2373           | 974    | 69       | Smiths Chip Thinly S/Cream&Onion 175g    | 5        | 15.0      |
| 4 | 43330 | 2         | 2426           | 1038   | 108      | Kettle Tortilla ChpsHny&Jlpno Chili 150g | 3        | 13.8      |

- 4. Check for the unique values and their counts in each column for the two DataFrames

```
grocery_purchase.nunique()
```

|                  | 0     |
|------------------|-------|
| LYLTY_CARD_NBR   | 72637 |
| LIFESTAGE        | 7     |
| PREMIUM_CUSTOMER | 3     |

dtype: int64

grocery\_transaction.nunique()

|                | 0      |
|----------------|--------|
| DATE           | 364    |
| STORE_NBR      | 272    |
| LYLTY_CARD_NBR | 72637  |
| TXN_ID         | 263127 |
| PROD_NBR       | 114    |
| PROD_NAME      | 114    |
| PROD_QTY       | 6      |
| TOT_SALES      | 112    |

dtype: int64

grocery transaction has at least two unique values

## 5. Check for duplicates in the two DataFrames

grocery\_purchase[grocery\_purchase.duplicated()]

LYLTY\_CARD\_NBR LIFESTAGE PREMIUM\_CUSTOMER

## No duplicate row in grocery\_purchase

grocery\_transaction[grocery\_transaction.duplicated()]

|        | DATE  | STORE_NBR | LYLTY_CARD_NBR | TXN_ID | PROD_NBR | PROD_NAME                            | PROD_QTY | TOT_SALES |
|--------|-------|-----------|----------------|--------|----------|--------------------------------------|----------|-----------|
| 124845 | 43374 | 107       | 107024         | 108462 | 45       | Smiths Thinly Cut Roast Chicken 175g | 2        | 6.0       |

## We have duplicate row in grocery\_transaction

```
### Remove duplicates from Pandas DataFrame-grocery_transaction
```

```
grocery_transaction.drop_duplicates()
```

|        | DATE  | STORE_NBR | LYLTY_CARD_NBR | TXN_ID | PROD_NBR | PROD_NAME                                | PROD_QTY | TOT_SALES |
|--------|-------|-----------|----------------|--------|----------|------------------------------------------|----------|-----------|
| 0      | 43390 | 1         | 1000           | 1      | 5        | Natural Chip Compny SeaSalt175g          | 2        | 6.0       |
| 1      | 43599 | 1         | 1307           | 348    | 66       | CCs Nacho Cheese 175g                    | 3        | 6.3       |
| 2      | 43605 | 1         | 1343           | 383    | 61       | Smiths Crinkle Cut Chips Chicken 170g    | 2        | 2.9       |
| 3      | 43329 | 2         | 2373           | 974    | 69       | Smiths Chip Thinly S/Cream&Onion 175g    | 5        | 15.0      |
| 4      | 43330 | 2         | 2426           | 1038   | 108      | Kettle Tortilla ChpsHny&Jlpno Chili 150g | 3        | 13.8      |
| ...    | ...   | ...       | ...            | ...    | ...      | ...                                      | ...      | ...       |
| 264831 | 43533 | 272       | 272319         | 270088 | 89       | Kettle Sweet Chilli And Sour Cream 175g  | 2        | 10.8      |
| 264832 | 43325 | 272       | 272358         | 270154 | 74       | Tostitos Splash Of Lime 175g             | 1        | 4.4       |
| 264833 | 43410 | 272       | 272379         | 270187 | 51       | Doritos Mexicana 170g                    | 2        | 8.8       |
| 264834 | 43461 | 272       | 272379         | 270188 | 42       | Doritos Corn Chip Mexican Jalapeno 150g  | 2        | 7.8       |
| 264835 | 43365 | 272       | 272380         | 270189 | 74       | Tostitos Splash Of Lime 175g             | 2        | 8.8       |

264835 rows × 8 columns

```
grocery_transaction.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 264836 entries, 0 to 264835
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   DATE             264836 non-null  int64
1   STORE_NBR        264836 non-null  int64
2   LYLTY_CARD_NBR   264836 non-null  int64
3   TXN_ID           264836 non-null  int64
4   PROD_NBR         264836 non-null  int64
5   PROD_NAME        264836 non-null  object
6   PROD_QTY         264836 non-null  int64
7   TOT_SALES        264836 non-null  float64
dtypes: float64(1), int64(6), object(1)
memory usage: 16.2+ MB
```

- ✓ .6 Check for missing data by columns in `grocery_purchase` using the `isna` method

```
grocery_purchase.isna().sum()
```

```
dtype: int64
```

|                  | 0 |
|------------------|---|
| LYLTY_CARD_NBR   | 0 |
| LIFESTAGE        | 0 |
| PREMIUM_CUSTOMER | 0 |

No missing data in this dataset

```
grocery_transaction.isna().sum()
```

```
dtype: int64
```

|                | 0 |
|----------------|---|
| DATE           | 0 |
| STORE_NBR      | 0 |
| LYLTY_CARD_NBR | 0 |
| TXN_ID         | 0 |
| PROD_NBR       | 0 |
| PROD_NAME      | 0 |
| PROD_QTY       | 0 |
| TOT_SALES      | 0 |

No missing data in this dataset

7. Check for outliers in `grocery_purchase` using the `describe` method, then look at the histograms of the suspicious columns

```
grocery_purchase.describe()
```

|              | LYLTY_CARD_NBR |
|--------------|----------------|
| <b>count</b> | 7.263700e+04   |
| <b>mean</b>  | 1.361859e+05   |
| <b>std</b>   | 8.989293e+04   |
| <b>min</b>   | 1.000000e+03   |
| <b>25%</b>   | 6.620200e+04   |
| <b>50%</b>   | 1.340400e+05   |
| <b>75%</b>   | 2.033750e+05   |
| <b>max</b>   | 2.373711e+06   |

The DataFrame `grocery_purchase` doesn't have any columns that could have outliers, so we are not exploring it.

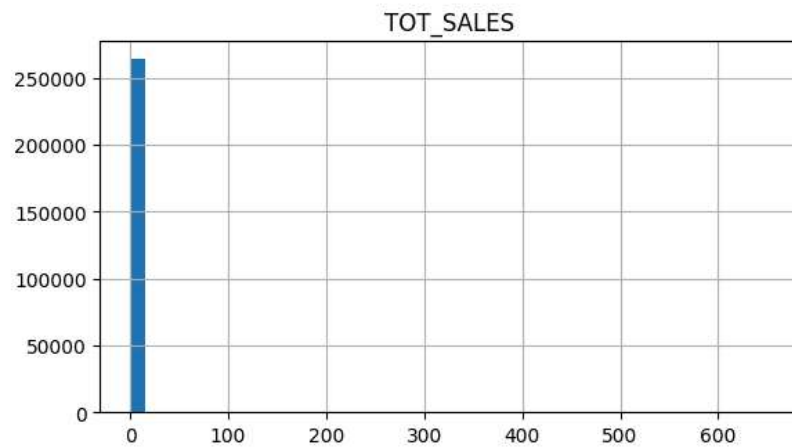
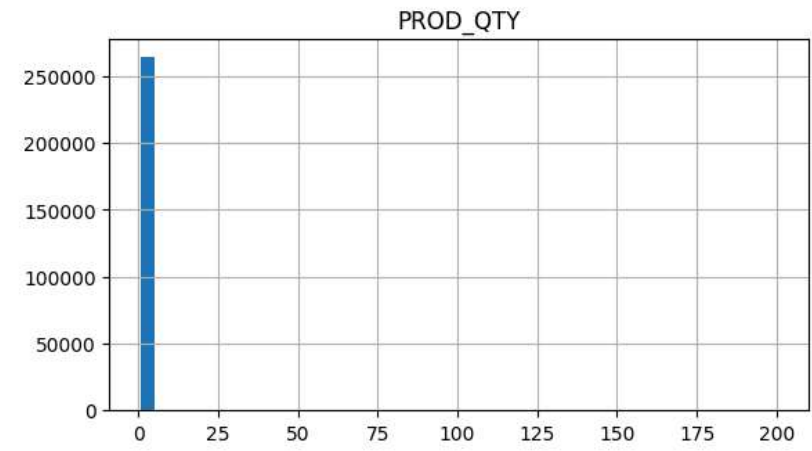
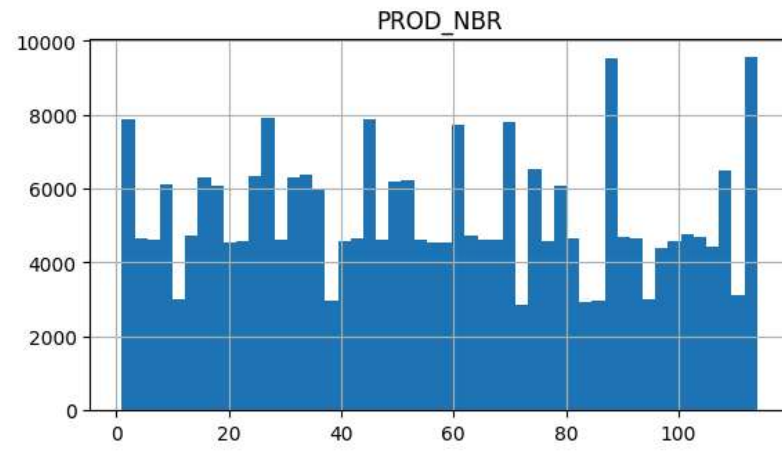
8. Check for outliers in `grocery_transaction` using the `describe` method, then look at the histograms of the suspicious columns

```
grocery_transaction.describe()
```

|              | DATE          | STORE_NBR     | LYLTY_CARD_NBR | TXN_ID       | PROD_NBR      | PROD_QTY      | TOT_SALES     |
|--------------|---------------|---------------|----------------|--------------|---------------|---------------|---------------|
| <b>count</b> | 264836.000000 | 264836.000000 | 2.648360e+05   | 2.648360e+05 | 264836.000000 | 264836.000000 | 264836.000000 |
| <b>mean</b>  | 43464.036260  | 135.08011     | 1.355495e+05   | 1.351583e+05 | 56.583157     | 1.907309      | 7.304200      |
| <b>std</b>   | 105.389282    | 76.78418      | 8.057998e+04   | 7.813303e+04 | 32.826638     | 0.643654      | 3.083226      |
| <b>min</b>   | 43282.000000  | 1.00000       | 1.000000e+03   | 1.000000e+00 | 1.000000      | 1.000000      | 1.500000      |
| <b>25%</b>   | 43373.000000  | 70.00000      | 7.002100e+04   | 6.760150e+04 | 28.000000     | 2.000000      | 5.400000      |
| <b>50%</b>   | 43464.000000  | 130.00000     | 1.303575e+05   | 1.351375e+05 | 56.000000     | 2.000000      | 7.400000      |
| <b>75%</b>   | 43555.000000  | 203.00000     | 2.030942e+05   | 2.027012e+05 | 85.000000     | 2.000000      | 9.200000      |
| <b>max</b>   | 43646.000000  | 272.00000     | 2.373711e+06   | 2.415841e+06 | 114.000000    | 200.000000    | 650.000000    |

```
grocery_transaction[['PROD_NBR', 'PROD_QTY', 'TOT_SALES']].hist(bins=50, figsize=(15,8))
```

```
array([[<Axes: title={'center': 'PROD_NBR'}>,
        <Axes: title={'center': 'PROD_QTY'}>],
       [<Axes: title={'center': 'TOT_SALES'}>, <Axes: >]], dtype=object)
```



```
grocery_transaction[grocery_transaction['PROD_NBR'] > 100]
```

|        | DATE  | STORE_NBR | LYLTY_CARD_NBR | TXN_ID | PROD_NBR | PROD_NAME                                | PROD_QTY | TOT_SALES |
|--------|-------|-----------|----------------|--------|----------|------------------------------------------|----------|-----------|
| 4      | 43330 | 2         | 2426           | 1038   | 108      | Kettle Tortilla ChpsHny&Jlpno Chili 150g | 3        | 13.8      |
| 11     | 43332 | 8         | 8294           | 8221   | 114      | Kettle Sensations Siracha Lime 150g      | 5        | 23.0      |
| 24     | 43327 | 38        | 38142          | 34181  | 108      | Kettle Tortilla ChpsHny&Jlpno Chili 150g | 2        | 9.2       |
| 26     | 43331 | 39        | 39167          | 35638  | 111      | Smiths Chip Thinly Cut Original 175g     | 2        | 6.0       |
| 39     | 43330 | 55        | 55072          | 48878  | 107      | Smiths Crinkle Cut French OnionDip 150g  | 1        | 2.6       |
| ...    | ...   | ...       | ...            | ...    | ...      | ...                                      | ...      | ...       |
| 264748 | 43284 | 268       | 268228         | 264647 | 102      | Kettle Mozzarella Basil & Pesto 175g     | 1        | 5.4       |
| 264752 | 43311 | 268       | 268365         | 264802 | 103      | RRD Steak & Chimuchurri 150g             | 1        | 2.7       |
| 264759 | 43499 | 269       | 269125         | 265775 | 107      | Smiths Crinkle Cut French OnionDip 150g  | 2        | 5.2       |
| 264784 | 43308 | 270       | 270002         | 266605 | 113      | Twisties Chicken270g                     | 2        | 9.2       |
| 264827 | 43340 | 272       | 272197         | 269911 | 104      | Infuzions Thai SweetChili PotatoMix 110g | 2        | 7.6       |

33044 rows × 8 columns

```
grocery_transaction.loc[grocery_transaction['PROD_NBR'] > 100, 'PROD_NAME'].value_counts()
```

|                                          | count |
|------------------------------------------|-------|
| PROD_NAME                                |       |
| Kettle Mozzarella Basil & Pesto 175g     | 3304  |
| Kettle Tortilla ChpsHny&Jlpno Chili 150g | 3296  |
| Tyrrells Crisps Ched & Chives 165g       | 3268  |
| Infuzions Thai SweetChili PotatoMix 110g | 3242  |
| Pringles Barbeque 134g                   | 3210  |
| Twisties Chicken270g                     | 3170  |
| Kettle Sensations Siracha Lime 150g      | 3127  |
| Smiths Chip Thinly Cut Original 175g     | 1614  |
| Woolworths Cheese Rings 190g             | 1516  |
| WW Original Corn Chips 200g              | 1495  |
| Natural ChipCo Hony Soy Chckn175g        | 1460  |
| RRD Steak & Chimuchurri 150g             | 1455  |
| Doritos Salsa Medium 300g                | 1449  |
| Smiths Crinkle Cut French OnionDip 150g  | 1438  |

dtype: int64

```
grocery_transaction[grocery_transaction['PROD_QTY'] > 0]
```

|        | DATE  | STORE_NBR | LYLTY_CARD_NBR | TXN_ID | PROD_NBR | PROD_NAME                                | PROD_QTY | TOT_SALES |
|--------|-------|-----------|----------------|--------|----------|------------------------------------------|----------|-----------|
| 0      | 43390 | 1         | 1000           | 1      | 5        | Natural Chip Compny SeaSalt175g          | 2        | 6,0       |
| 1      | 43599 | 1         | 1307           | 348    | 66       | CCs Nacho Cheese 175g                    | 3        | 6,3       |
| 2      | 43605 | 1         | 1343           | 383    | 61       | Smiths Crinkle Cut Chips Chicken 170g    | 2        | 2,9       |
| 3      | 43329 | 2         | 2373           | 974    | 69       | Smiths Chip Thinly S/Cream&Onion 175g    | 5        | 15,0      |
| 4      | 43330 | 2         | 2426           | 1038   | 108      | Kettle Tortilla ChpsHny&Jlpno Chili 150g | 3        | 13,8      |
| ...    | ...   | ...       | ...            | ...    | ...      | ...                                      | ...      | ...       |
| 264831 | 43533 | 272       | 272319         | 270088 | 89       | Kettle Sweet Chilli And Sour Cream 175g  | 2        | 10,8      |
| 264832 | 43325 | 272       | 272358         | 270154 | 74       | Tostitos Splash Of Lime 175g             | 1        | 4,4       |
| 264833 | 43410 | 272       | 272379         | 270187 | 51       | Doritos Mexicana 170g                    | 2        | 8,8       |
| 264834 | 43461 | 272       | 272379         | 270188 | 42       | Doritos Corn Chip Mexican Jalapeno 150g  | 2        | 7,8       |
| 264835 | 43365 | 272       | 272380         | 270189 | 74       | Tostitos Splash Of Lime 175g             | 2        | 8,8       |

264836 rows × 8 columns

```
grocery_transaction.loc[grocery_transaction['PROD_QTY'] > 0, 'PROD_QTY'].value_counts()
```

|          | count  |
|----------|--------|
| PROD_QTY |        |
| 2        | 236039 |
| 1        | 27518  |
| 5        | 450    |
| 3        | 430    |
| 4        | 397    |
| 200      | 2      |

dtype: int64

```
grocery_transaction[grocery_transaction['TOT_SALES'] > 0]
```

|        | DATE  | STORE_NBR | LYLTY_CARD_NBR | TXN_ID | PROD_NBR | PROD_NAME                                | PROD_QTY | TOT_SALES |
|--------|-------|-----------|----------------|--------|----------|------------------------------------------|----------|-----------|
| 0      | 43390 | 1         | 1000           | 1      | 5        | Natural Chip Compny SeaSalt175g          | 2        | 6.0       |
| 1      | 43599 | 1         | 1307           | 348    | 66       | CCs Nacho Cheese 175g                    | 3        | 6.3       |
| 2      | 43605 | 1         | 1343           | 383    | 61       | Smiths Crinkle Cut Chips Chicken 170g    | 2        | 2.9       |
| 3      | 43329 | 2         | 2373           | 974    | 69       | Smiths Chip Thinly S/Cream&Onion 175g    | 5        | 15.0      |
| 4      | 43330 | 2         | 2426           | 1038   | 108      | Kettle Tortilla ChpsHny&Jlpno Chili 150g | 3        | 13.8      |
| ...    | ...   | ...       | ...            | ...    | ...      | ...                                      | ...      | ...       |
| 264831 | 43533 | 272       | 272319         | 270088 | 89       | Kettle Sweet Chilli And Sour Cream 175g  | 2        | 10.8      |
| 264832 | 43325 | 272       | 272358         | 270154 | 74       | Tostitos Splash Of Lime 175g             | 1        | 4.4       |
| 264833 | 43410 | 272       | 272379         | 270187 | 51       | Doritos Mexicana 170g                    | 2        | 8.8       |
| 264834 | 43461 | 272       | 272379         | 270188 | 42       | Doritos Corn Chip Mexican Jalapeno 150g  | 2        | 7.8       |
| 264835 | 43365 | 272       | 272380         | 270189 | 74       | Tostitos Splash Of Lime 175g             | 2        | 8.8       |

264836 rows × 8 columns

```
grocery_transaction.loc[grocery_transaction['TOT_SALES'] > 0, 'TOT_SALES'].value_counts()
```

|           | count |
|-----------|-------|
| TOT_SALES |       |
| 9.2       | 22821 |
| 7.4       | 22513 |
| 6.0       | 20798 |
| 7.6       | 20212 |
| 8.8       | 19900 |
| ...       | ...   |
| 9.3       | 3     |
| 6.9       | 3     |
| 12.4      | 2     |
| 650.0     | 2     |
| 11.2      | 2     |

112 rows × 1 columns

dtype: int64

9. Convert column `DATE` in `grocery_transaction` into datetime dtypes

```
grocery_transaction['DATE']=pd.to_datetime(grocery_transaction['DATE'])
```

```
grocery_transaction.dtypes
```

```
0
```

```
DATE      datetime64[ns]
```